

# 10 Effect of Stretch Training on Functional Performance

## Static Stretching

While acute prolonged static stretching can lead to impairments, there are physiological rationales why chronic static stretching could improve performance. Static stretch training can enhance  $\text{Ca}^{+}$  within the neuromuscular junction (37), decrease muscle stiffness (35), and proliferate sarcomeres in series (9, 39). Increased neuromuscular junction  $\text{Ca}^{+}$  should enhance neurotransmitter release. Decreased musculotendinous stiffness or increased compliance could reduce the resistance to movement improving movement and energy efficiency. Energy efficiency would be directly related to a training-related decrease in hysteresis (18). Hysteresis is related to tissue viscoelastic changes affecting heat loss with stretching (14). Thus training-induced changes in viscoelasticity leading to decreased hysteresis (the return to the original state is delayed) would help retain rather than dissipate energy through heat loss (18). Furthermore, increased musculotendinous unit compliance could positively affect the stretch-shortening cycle (SSC) by enhancing the ability to store elastic energy (18, 35). This compliance-induced elastic storage enhancement would be more apparent with longer duration eccentric to concentric transitions such as a rebound chest press (35) but might not have such a positive impact on very short SSC activities like sprinting where the athlete might transition from foot strike to take-off in less than 100 ms.

A higher number of sarcomeres in series could allow muscle to produce more force or torque at longer muscle lengths and change the optimal force angle (8). It is reported that many of the musculotendinous injuries occur at extended muscle lengths when the muscle exerts lower forces and cannot protect itself as well against environmental stressors to as great an extent (4). Another advantage is that the number of sarcomeres in series is related to muscle contraction velocity (15). As a reminder though, increases in sarcomere number with stretching has been primarily verified in animal studies but not human in-vivo studies.

Although these proposed physiological adaptations should provide physical benefits, does the practical research concur? A well cited review by Shrier (24) reported that regular stretching would improve most athletic performances. Another systematic review (18) showed that only half of the stretch training

articles (14/28) improved muscle performance. There was very little effect upon static or isometric contraction measures. Muscle performance improvements in these studies were generally dynamic activities such as jumps, isokinetic eccentric and concentric torque, rebound bench press, and plantar flexors 1RM. However, even with dynamic activities the results were not consistent. Similar to the Behm et al. (4) review that suggested a major advantage of stretching would be enhanced force outputs at longer muscle lengths, the Medeiros and Lima review (18) found a number of studies reporting enhanced eccentric peak torque. As eccentric contractions are ubiquitous in all activities but especially important in SSC activities, as they help to potentiate or augment the concentric contraction (6, 7), a flexibility training-induced improvement in eccentric strength could be of vital importance. As you should suspect by now, none of these statements ever get unanimous support.

For example, while it is safe to state that almost every chronic stretch training programme increases range of motion (ROM) (13), there are a number of studies that reported no change in evoked contraction speed (8 weeks) (1), sprint time and vertical jump height (4–6 weeks) (3, 11, 22), countermovement jump height (nine weeks in pre-adolescent athletes) (10), drop jump height (four and six weeks) (19, 38), dynamic knee flexion strength (six weeks) (5), plantar flexors 1RM (ten weeks) (20), or maximum voluntary isometric contraction force (four, six, and 12 weeks) (17, 19, 22, 23). Some chronic stretch training studies actually show deficits, with for example decreases in eccentric peak torque and triple hop distance (ten sessions) (2). Just to increase the confusion, there are a number of studies demonstrating stretch training-induced improvements in hamstrings peak torque and countermovement jump height (eight weeks) (16), vertical jump height (four weeks: squat and countermovement jump) (21), dynamic postural stability (six weeks) (22) and MVIC force (five weeks) (36). Lima et al. (16) reported greater effect sizes (larger magnitude improvements) with an 8 week periodized stretch training programme versus non-periodized for ROM and performance. Why is there such diversity in these results? There was very little consistency in the stretching prescriptions in these studies. Prescriptions ranged from session durations of 2x30 seconds, 3x30s, 5x30s, 3x60s, 5x45s, or 2 minutes, weekly frequencies of one, three, four, or five days/week and training programme durations of ten sessions, 15 days, four, six, eight, ten, or 12 weeks. In these studies, we cannot see a trend for improved performance with longer training durations or higher weekly frequencies or even longer session durations. However, compared to some recent research showing stretch-induced increases in muscle strength and hypertrophy (26, 27, 28, 29, 30, 31, 32, 33, 34) these cited session durations are relatively short (<4 minutes) and often only three days per week. A recent review of chronic stretching effects on jump and sprint performance demonstrated a significant but trivial magnitude increase in jump height but no significant effect on sprint time or speed. We summarized and theorized that static stretching was not a sufficient stimulus to substantially enhance jump and sprint performance, possibly owing to small weekly training volumes or lack of intensity.

Table 10.1 Dynamic Stretching FITT table

| <i>FITT Recommendations</i> |                                                                                                                                                                                                                                                 |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Frequency                   | <ul style="list-style-type: none"> <li>• Daily (<math>\geq 5</math> per week provide optimal benefits, but improvements are still seen at the rate of twice per week)</li> </ul>                                                                |
| Intensity                   | <ul style="list-style-type: none"> <li>• Full range of motion (dynamic stretching: submaximal limb velocity)</li> </ul>                                                                                                                         |
| Time                        | <ul style="list-style-type: none"> <li>• 3–6 sets of 30–90 seconds with 15s rest periods between sets</li> </ul>                                                                                                                                |
| Type                        | <ul style="list-style-type: none"> <li>• Dynamic stretching for all major muscle-tendon units is recommended</li> <li>• Dynamic and Ballistic (higher-velocity movements typically with bouncing actions at the end ROM) flexibility</li> </ul> |

The next chapter will highlight the more consistently positive results on muscle strength gains when more prolonged static stretching durations (10–60 minutes per muscle group) are implemented into nearly daily (e.g., 5–6 days per week) stretch training programmes.

## Dynamic Stretch Training

When dynamic stretching was incorporated into the daily warm-up of wrestlers for 4 weeks there were improvements in power, strength, muscular endurance, anaerobic capacity, and agility performance (12). Similarly, an eight-week programme involving either active (not staying in one spot) or static (staying in one location) dynamic stretch training improved both flexibility and jump parameters but not sprint performance (25). However, there are also reports of no changes of eccentric peak torque and triple hop test distance (3x week for ten sessions) (2).

## Summary

In contrast to the reported performance impairments with acute bouts of prolonged static stretching, there are many physiological adaptations with static stretch training that should enhance performance. However, the evidence from these low volume or duration static stretch training studies is contradictory. The few dynamic stretch studies do show performance improvements but not with every measure.

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